



Faecal microbiota transplantation in patients with haematological malignancies undergoing cellular therapies: from translational research to routine clinical practice

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The effect of the gut microbiota on patients' outcomes after allogeneic haematopoietic cell transplantation (HCT) is now well established. In particular, gut microbiota dysbiosis has been associated with acute graft-versus-host disease (GVHD). Furthermore, increasing data also suggest an effect of the gut microbiota on outcome after autologous HCT and CAR T cells. In fact, the bacterial gut microbiota interplays with the immune system and contributes to immunological complication and antitumour response to treatment. Therefore, faecal microbiota transplantation has been evaluated in patients with haematological malignancies for various indications, including *Clostridioides difficile* infection, eradication of multidrug-resistant bacteria, and steroid refractory acute GVHD. In addition, use of prophylactic faecal microbiota transplantation to restore the gut microbiota and improve patients' outcomes is being developed in the setting of allogeneic HCT, but also probably very soon in patients receiving autologous HCT or CAR T cells.

Introduction

The microbiota, the community of microorganisms colonising humans and their collective genomes (the microbiome), is closely associated with health and disease. The human gastrointestinal tract encompasses around 100 trillion microorganisms, most of them being obligate anaerobic bacteria, but also aerobic bacteria, viruses, fungi, and protozoa.¹ Although the human genome consists of about 23 000 genes, each person has about 600 000 microbial genes,^{2,3} and one study identified more than 22 million non-redundant genes in the human gut microbiome.⁴ Therefore, although two main phyla, *Bacteroidetes* and *Firmicutes*, constitute more than 90% of the bacterial community in a healthy person,^{2,3} microbiota composition is highly diverse and variable at low taxonomic levels (genus and species) between individuals.⁵

The microbiota and host coevolve from birth, leading to a host–microbiome symbiosis in healthy individuals.¹ Therefore, the gut microbiome contributes to health through several essential functions in digestive metabolism, barrier function, prevention of the proliferation of environmental microbes including pathogens, stimulation of natural defenses, development of the intestinal wall, and immune system stimulation.⁶

Alteration of the host–microbiome symbiosis is associated with disruption of balance of the gut microbiota composition, known as dysbiosis, with alteration of the gut barrier (most often permeability) that leads to a pathological condition with uncontrolled local immune response and potentially systemic inflammation.⁷ This symbiosis can be disrupted by alterations because of nutrition or iatrogenic condition. In particular, patients with haematological malignancies have well established factors of dysbiosis, with chemotherapy and broad-spectrum antibiotics profoundly altering the gut microbiota composition.^{6,8} This alteration is characterised by an overall decrease in microbial richness and diversity with a loss of beneficial bacteria that support the host

defence, such as *Firmicutes* and a dominance of usually subdominant bacterial species, mostly pathogens or pathobionts, essentially Gram-negative proinflammatory microbes.⁷ The alteration increases intestinal permeability and inflammation, leading to the vicious circle of oxidative stress and further aggravation and alteration of the gut microbiota.^{9,10} These changes result in a durably altered symbiosis as seen in chronic diseases including infection, cancer, inflammatory and metabolic disease, neurological disease, and mental illness.¹¹ Nevertheless, altered symbiosis is potentially reversible. Providing adequate treatment targets the main trigger: intestinal permeability, inflammation, oxidative stress, and low richness microbiota.¹ Therefore, microbiotherapy with faecal microbiota transplantation (FMT) appears central in the treatment options to restore microbiota richness, and correct microbiota dysbiosis and altered host–microbiome symbiosis associated with chronic diseases.

Effect of gut microbiota on patients' outcomes after allogeneic haematopoietic cell transplantation (HCT)

In patients with haematological malignancies, a link between gut microbiota diversity and patients' outcomes was first established in patients receiving allogeneic HCT. In 2014, Taur and colleagues¹² showed that overall survival decreased in a cohort of 80 patients with lower gut microbiota diversity at engraftment compared with patients with higher diversity. This finding was confirmed in a large multicenter study published in 2020, which included 1362 patients treated in four centres (two in the USA, one in Germany, and one in Japan): higher diversity of intestinal microbiota at engraftment was associated with a lower risk of death than for patients with a lower diversity (adjusted hazard ratio [HR] 0.71; 95% CI, 0.55–0.92 for cohort 1; 0.49; 0.27–0.90 for cohort 2).¹³ Furthermore, they found that gut microbiota diversity before the start of the conditioning regimen was lower than for healthy patients, and patients with higher

microbiota diversity at this time point were at a lower risk of mortality after allogeneic HCT than were patients with lower diversity (0.41; 0.24–0.71).

Importantly, it was shown in a large group of patients (n=1325) that in addition to low microbial diversity, intestinal domination is frequent after allogeneic HCT (65% of patients) mainly related to *Enterococcus faecium*.¹⁴ The authors found that faecal domination with *Enterococcus* in the early post-transplantation period (0–21 days) was associated not only with a reduced overall survival but also with an increased mortality related to graft-versus-host disease (GVHD), and an increased risk of moderate-to-severe acute GVHD. By contrast, Jenq and colleagues¹⁵ found that increased numbers of bacteria belonging to the genus *Blautia* (order Clostridiales) were associated with reduced GVHD lethality and improved overall survival. Of note, loss of *Blautia* was associated with antibiotics that inhibit anaerobic bacteria and use of total parenteral nutrition for a long duration. *Blautia* is a butyrate producer, a short-chain fatty acid that modulates intestinal epithelial cell damage after allogeneic HCT and mitigates GVHD in a mouse model.¹⁶ In addition, at the onset of acute GVHD in humans, there are some substantial variations in microbiota-derived metabolites, especially in aryl hydrocarbon receptor ligands, bile acids, and plasmalogens.¹⁷ Notably, in 2022, it was shown that alloreactive T cells deficient in the main receptor of butyrate, GPR109A, induce less GVHD in a mouse model.¹⁸ In chronic GVHD, although α -diversity patterns are similar over time in selected patients with chronic GVHD and control participants, circulating concentrations of the microbe-derived short-chain fatty acids, propionate and butyrate, in day 100 plasma samples were lower in patients who developed chronic GVHD than in those patients who remained free of this complication.¹⁹ Overall, this outcome suggests that gut microbiota dysbiosis induces major change in circulating metabolites in recipients, which influence allogeneic immune cell reactivity and GVHD.

Shono and colleagues²⁰ looked more closely at the effect of antibiotics with a wide antianaerobic spectrum in a cohort of 847 patients with allogeneic HCT and found that treatment of neutropenic fever with imipenem–cilastatin and piperacillin–tazobactam was associated with perturbation of gut microbial composition and increased GVHD-related mortality at 5 years. The detrimental antibiotic effect was further confirmed by Weber and colleagues²¹ who showed in a cohort of 621 patients with allogeneic HCT that early antibiotic exposure (before day 0 of allogeneic HCT), was associated with lower urinary 3-indoxyl sulphate concentrations (mean 5.5 μ mol/mmol creatinine; SD 13.6; $p<0.001$), a decrease in faecal abundance of commensal Clostridiales (0.74%; $p=0.03$), and higher transplantation-related mortality (34%; 79/236) than for patients who started antibiotics after allogeneic HCT (21%; 62/297; $p=0.001$) or did not receive antibiotics (7%; 6/88; $p<0.001$).²¹

The link between gut microbiota diversity and risk of relapse or progression of the underlying malignancy has also been investigated.²² Compared with a lower abundance, a higher abundance of a bacterial group composed mostly of *Eubacterium limosum* (order Clostridiales) was associated with a decreased risk of relapse or progression of disease (HR 0.82 per 10-fold increase in abundance; 95% CI 0.71–0.95; $p=0.009$). *Eubacterium limosum* is involved in butyrate production, suggesting an immunomodulatory effect underlying the observed association.

Effect of gut microbiota in patients with haematological malignancies beyond allogeneic HCT

Autologous HCT

One large observational study included 534 adult patients who received an autologous HCT for lymphoma, myeloma, or amyloidosis at two centres in the USA.²³ Similar to what was reported in patients who received an allogeneic HCT, diversity of an early pretransplant faecal microbiota was lower in patients (median inverse Simpson index 9.39) than in the healthy control group (15.87; $p=0.003$) and decreased further during the course of transplantation. Furthermore, above-median faecal intestinal diversity in the periengraftment period was associated with decreased risk of death or progression in the patients (progression-free survival HR 0.46; 95% CI 0.26–0.82; $p=0.008$), after adjusting for disease and disease status. In addition, a small single centre study found that 28 patients who developed bacteraemia or bloodstream infection after autologous HCT for non-Hodgkin lymphoma exhibited decreased overall diversity and decreased abundance of certain taxa.²⁴ Another single centre study looked in more detail at the link between microbiota diversity and autologous HCT toxicity in 15 patients with multiple myeloma (n=14) or polyneuropathy, organomegaly, endocrinopathy, M protein, and skin lesions (known as POEMS) syndrome (n=1). They found that the baseline microbiome composition was associated with the incidence and severity of post-transplantation nausea, vomiting, and culture-negative neutropenic fever, and with the rate of neutrophil engraftment. They also observed an association between the nadir count of microbial communities and the development of regimen-related gastrointestinal toxicities commonly observed after exposure to high-dose melphalan.

Nevertheless, in the study by Khan and colleagues,²³ participants had only a few non-relapse mortality events and the mortality difference between patients with low versus high intestinal diversity in the periengraftment period was explained by a decreased progression-free survival in recipients with lower faecal diversity compared with recipients with higher diversity, suggesting an association between microbiota composition and antitumour response after autologous HCT.

In a murine model, the alkylating agent cyclophosphamide altered the composition of the microbiota in the small intestine and induced the translocation of selected species of Gram-positive bacteria into secondary lymphoid organs, leading to the generation of a specific subset of pathogenic T helper 17 (pTH17) cells.²⁵ Tumour-bearing mice that were germ-free or that had been treated with antibiotics to kill Gram-positive bacteria showed a reduction in pTH17 response, and their tumours were resistant to cyclophosphamide, whereas adoptive transfer of pTH17 cells partly restored the antitumour efficacy of cyclophosphamide. These results suggest an immune antitumour effect of cyclophosphamide, mediated by the gut microbiota.

CART-cell therapy

Smith and colleagues²⁶ evaluated the effect of antibiotic exposure on the gut microbiome in a retrospective cohort of 228 patients with non-Hodgkin lymphoma (n=137) or acute lymphocytic leukaemia (n=91) treated with anti-CD19 CAR T-cell therapy. They found that exposure to antibiotics, in particular to the anaerobe-targeting antibiotics piperacillin-tazobactam, meropenem and imipenem-cilastatin in the 4 weeks before therapy was associated with worse overall survival and increased neurotoxicity than for patients who did not receive those antibiotics. Furthermore, the faecal microbiome was analysed at baseline in a prospective cohort of 48 patients who received anti-CD19 CAR T-cell therapy. They found that the faecal microbiota was altered at baseline in patients who received CAR T cells, with a lower α -diversity and a different composition compared with healthy volunteers. They also identified species within the class Clostridia that were associated with complete response at day 100.

Similarly, a small study (published as a congress abstract) of 33 patients who received anti-CD19 CAR T-cell therapy for non-Hodgkin lymphoma or acute lymphocytic leukaemia showed that the use of anti-infective treatment up to 2 weeks before anti-CD19 CAR T-cell therapy was associated with a lower response rate than for patients who had been treated with antibiotics or antifungals, or both, after day 0 of CAR T-cell infusion.²⁷

Multiple myeloma and lymphoma

Faecal microbiota composition seems to be important for multiple myeloma disease control including patients who did not receive autologous HCT. Gut microbiota composition has been evaluated in 34 patients with multiple myeloma after completion of first-line treatment and before initiation of lenalidomide maintenance.²⁸ 14 patients received an autologous HCT (41%) as part of this first-line treatment. Bone marrow minimal residual disease (MRD) was assessed by flow cytometry at the same time. Although no differences in microbiota α -diversity were observed between patients with and

without MRD, the relative abundance of the butyrate-producing bacterium *Eubacterium hallii* was higher in the 16 patients without MRD than in the 18 patients with MRD. No association was observed between microbial relative abundance and autologous HCT history. In addition, the microbiota has also been involved in multiple myeloma promotion in a murine model.²⁹ *Prevotella heparinolytica* promotes the differentiation of Th17 cells colonising the gut and migrating to the bone marrow of transgenic Vk*MYC mice, where they favour progression of multiple myeloma in synergy with eosinophils. Overall, although these studies require further validation, and along with the improved disease-free survival in patients with a higher faecal intestinal diversity in the peri-engraftment phase after autologous HCT,²³ there is a strong rationale supporting a link between gut microbiota composition and antitumoural response in multiple myeloma.

In 2021, a study reported that the pretreatment gut microbiota in 51 patients with lymphoma had a distinct composition compared with 58 healthy controls ($p<0.001$).³⁰ Furthermore, gut microbiota composition also differed between patients with aggressive (n=8) or indolent lymphoma (n=18) ($p=0.01$). Finally, gut microbiota pretreatment diversity and composition were predictors of response to immunochemotherapy (eight responders vs six non-responders; $p=0.01$). Although these findings deserve confirmation in a larger multicentre study, they highlight the possible role of the gut microbiota in progression or development of lymphoma and response to treatment.

The concept of microbiota modulation

Based on these findings, there is a growing interest in gut microbiota modulation and several approaches are being developed. Firstly, dietary changes, such as fibre supplementation, or administration of prebiotics can change microbiota composition.^{31–33} Prebiotics are non-digestible food ingredients that stimulate growth and metabolism of beneficial bacteria in the gut.³⁴ Importantly, dietary changes and prebiotics are more likely to be effective if the symbiotic bacterial strains necessary to build a healthy microbiota are already present. When they are absent, as is frequently the case in patients with haematological malignancies who have received intensive chemotherapy and wide-spectrum antibiotics, the symbiotic bacterial strains will need to be administered in addition to the prebiotics.

Administration of probiotics is the second approach to modulate the gut microbiota. Probiotics are live microorganisms; either a single symbiotic bacterial strain or a consortium of selected strains can be administered. However, the gut microbiota consists of hundreds of microbial species, and development of a consortium of symbiotic bacterial strains is difficult.³⁵ In fact, no data exist on how different symbiotic bacterial strains support or reject each other in the gut microbiota.

Therefore, in the expectation of these next-generation probiotics, transfer of an entire faecal microbiota (FMT) is being studied in a large range of diseases.³⁵

FMT consists of the transfer of a complex community of microorganisms from a healthy donor to a recipient. Strict screening to rule out presence of pathogens in the healthy donor is imperative.³⁶ The purpose of FMT is to restore the beneficial function of a healthy microbiota and the immune system crosstalk. Effectiveness of FMT was first shown in patients with recurrent *Clostridioides* (formerly *Clostridium*) *difficile* infection (CDI) with around a 90% success rate,³⁷ and FMT was proven to be superior to conventional antibiotic administration in a randomised clinical trial.³⁸ Therefore, FMT is now the treatment for a second or further recurrence of CDI.³⁹ FMT is under investigation in a large range of diseases, including inflammatory bowel disease, metabolic syndrome, neuropsychiatric disorders, haematological malignancies, and response to chemoimmunotherapy for solid tumours.³⁵

Regarding safety, Kelly and colleagues⁴⁰ reported the use of FMT for CDI in a retrospective cohort of 80 immunocompromised patients. The overall complete response was 89%. None had infections related to FMT, but two patients developed unrelated infections and five had self-limited diarrheal illness in which no causal organism was identified. Nevertheless, concerns were later raised after the report of drug-resistant *Escherichia coli* bacteraemia transmitted by FMT in two immunosuppressed patients (one patient with cirrhosis of the liver, one patient with a history of allogeneic HCT) lead to the death of one patient.⁴¹ Both cases were linked to the same stool donor. This concern led to a strengthening of FMT donor screening by the health authorities' agencies and international recommendations have been issued for FMT donor screening.³⁶

The clinical outcome of FMT can be affected by the FMT approach, in particular the route of administration and the type of donor. FMT can be administered via the lower gastrointestinal (colonoscopy or rectal enema) or upper gastrointestinal tract (endoscopy, nasoduodenal or nasojejunal tube, or capsule).⁴² There is no general agreement regarding the superiority of one route over another. Therefore, the choice of delivery route of FMT preparations should be based on local preference and patient characteristics if applicable.³⁶ Regarding donors, given the complexity, time, and cost of screening, universal donors are now preferred over patient-selected donors, with the constitution of a stool bank with frozen FMT material readily available. Importantly, frozen FMT was proven to be equivalent to fresh FMT in a randomised clinical trial and in a meta-analysis;^{43–45} therefore, frozen FMT is now the preferred choice.

We should also distinguish single donors from a pool of donors. Several studies suggest that FMT efficacy might be dependent on the diversity and composition of the microbiota of the stool donor, leading to the concept of an

FMT super donor effect.^{46–49} This effect was first reported in a study that evaluated FMT versus placebo in 75 patients with ulcerative colitis.⁴⁷ FMT (24%) was superior to placebo (5%), with seven (78%) of nine patients in remission in the FMT group receiving FMT from the same donor. To homogenise the donor-dependent effect, FMT can be used from a pool of several donors. A donor pool was proven to have higher microbial diversity than a single donor based on operational taxonomic unit and phylogenetic diversity measures.⁴⁸ One meta-analysis found a higher efficacy of pooled donor stools than for single donor stool in patients with ulcerative colitis in clinical remission.⁴⁹

Many microbiota-derived mediators with potentially beneficial responses in the host have been identified and are candidate treatments.³⁵ In the setting of allogeneic HCT, the use of butyrate as a treatment should be investigated given its potential protective effect on GVHD.^{16–19}

In the setting of patients with haematological malignancies, use of antibiotics should be reconsidered with regards to decreasing their burden on gut microbiota diversity. Rifaximin, an antibiotic that maintains the commensal microbiota and exerts anti-inflammatory activity,⁵⁰ seems a promising alternative to ciprofloxacin and metronidazole for gut decontamination after allogeneic HCT. In a retrospective study, use of rifaximin (n=194) was associated with a lower 1-year transplantation-related mortality and higher overall survival than for patients who received ciprofloxacin and metronidazole (n=200).⁵¹ Furthermore, use of rifaximin correlated with lower enterococcal positivity (6.9% [4/58] for rifaximin vs 21.9% [7/32] for ciprofloxacin and metronidazole; p=0.05) and higher urinary 3-indoxyl sulfate concentrations, indicating a higher intestinal diversity. Regarding use of antibiotics for febrile neutropenia, the European Conference on Infection in Leukemia published some guidelines to help decrease antibiotic burden, and therefore their detrimental effect on gut microbiota diversity.⁵² Carbapenem, in particular, should be avoided as first-line treatment for febrile neutropenia in patients without a particular risk factor. Furthermore, discontinuation of antibiotics should be reassessed after 72 h or later in patients with neutropenic fever of unknown origin who are haemodynamically stable since presentation and afebrile for at least 48 h.

FMT in immunocompromised patients with haematological diseases and infectious complications

In patients with haematological disease, the first report of FMT in a patient after allogeneic HCT was published in 2012, in a patient with a severe fulminant CDI. A single FMT with stools from the patient's husband was administered through a nasojejunal tube and resulted in resolution of CDI.⁵³ Thereafter, several case reports and a

	Indication or population	Number of patients and transplant type	Administration route	Study type	Donor relation	Number of FMT procedures	Response
CDI							
Neemann et al (2012) ⁵³	Severe fulminant CDI	1 allogeneic HCT	Nasojunal tube	Case report	Husband	1	Resolution of CDI
de Castro et al (2015) ⁵⁴	Recurrent CDI	1 autologous HCT	Push enteroscopy	Case report	Third party donor	1	No recurrence of CDI
Mittal et al (2015) ⁵⁵	Recurrent CDI	1 autologous HCT	Enema	Case report	Third party donor	2	No recurrence of CDI
Webb et al (2016) ⁵⁶	Recurrent CDI	7 autologous HCT	Nasojunal tube or colonoscopy	Retrospective, case series	Third party donor	8	6/7 had no recurrence of CDI
Moss et al (2017) ⁵⁷	Recurrent CDI	8 autologous or allogeneic HCT	Nasoduodenal tube	Retrospective, case series	Third party donor	8	7/8 had no recurrence of CDI
Bluestone et al (2018) ⁵⁸	Recurrent CDI/ pediatric	3 allogeneic HCT	Gastric tube or colonoscopy	Retrospective, case series	Relative or third party donor	8	1/3 had no recurrence of CDI
MRB eradication							
Bilinski et al (2017) ⁵⁹	MRB eradication	20 (n=8 allogeneic HCT; n=12 others haematological malignancies)	Nasoduodenal tube	Prospective	Third party donor	25	15/20 had MRB eradication
Innes et al (2017) ⁶⁰	MRB eradication	1 (before allogeneic HCT)	Nasogastric tube	Case report	Third party donor	1	MRB eradication
Battipaglia et al (2019) ⁶¹	MRB eradication	10 allogeneic HCT (n=4 before allogeneic HCT)	Enema or nasogastric tube	Retrospective	Third party donor	13 (n=9 after allogeneic HCT)	7/10 had MRB eradication
Su et al (2021) ⁶²	MRB eradication	1 (after allogeneic HCT)	Nasoduodenal tube	Case report	Third party donor	1	MRB eradication
No serious adverse events were reported. FMT=faecal microbiota transplantation. CDI=Clostridioides difficile infection. MRB=multidrug-resistant bacteria. HCT=haematopoietic cell transplantation.							
Table 1: FMT studies in HCT recipients for the treatment of CDI and MRB							

small series of FMT for recurrent CDI were reported in patients who had autologous HCT or allogeneic HCT (table 1). In particular, Moss and colleagues⁵⁷ reported results for eight patients (six allogeneic HCT and two autologous HCT), using frozen capsules from unrelated donors (15 capsules per patient) for recurrent CDI. FMT was well tolerated, and CDI resolved for all patients at 8 weeks with only one patient presenting with CDI recurrence 179 days post-FMT. In addition, 2 retrospective studies reported results of FMT for CDI recurrence in patients with malignancies. The first included 23 patients, 13 with haematological malignancies and ten with solid malignancies.⁶³ There was no serious adverse event, including infectious complication attributable to FMT. In all but three patients, diarrhoea resolved within 60 days and only two patients had recurrent CDI at 14 months and 22 months after FMT. The second retrospective study included 12 patients with solid malignancies and seven with haematological malignancies.⁶⁴ FMT from an unrelated donor was delivered to the cecum via regular colonoscopy. Overall, 16 (84%) of the 19 patients responded to FMT, whereas two patients had persistent diarrhoea and one patient had recurrent diarrhoea with a positive *C difficile* test within 4 weeks after FMT. In addition, two (11%) patients had a recurrent CDI after 1 year. Similar to other studies, there were no serious adverse events associated with FMT. According to the 2021 European Society of Clinical Microbiology and Infectious Diseases guidelines,³⁹ FMT is recommended for the treatment of second CDI recurrence and, based on the safety and results previously reported,^{57,63,64} immunocompromised patients (including

those with haematological malignancies and history of allogeneic HCT) should be considered for this treatment in this setting.

In addition to CDI, FMT has also been investigated for eradication of multidrug-resistant bacteria (table 1). Bilinski and colleagues⁶⁵ reported first results in 20 immunosuppressed patients including eight patients with a history of allogeneic HCT. 25 fresh FMT samples originating from three different donors were administered to the 20 patients by nasoduodenal tube. Three patients received two FMT and one patient received three FMT for persistence or recurrence of multidrug-resistant bacteria after previous FMT. Overall, eradication of multidrug-resistant bacteria was achieved in 15 (60%) of 25 FMT procedures. Similarly, Battipaglia and colleagues⁶¹ reported efficacy of FMT for eradication of multidrug-resistant bacteria in ten patients before (n=4) or after (n=6) allogeneic HCT. Fresh (n=2) or frozen (n=8) treatments from related or unrelated donors were administered by enema in all but one patient who received the FMT through a nasogastric tube. Three patients received a second FMT in the absence of response after the first procedure. Seven (70%) patients achieved complete decolonisation, which was persistent in six (60%) patients with a median follow-up of 13 months (range 4–40). Of note, one (10%) patient presented with constipation during the first 5 days after FMT, and two (20%) patients had grade 1 diarrhoea. No patient presented with serious adverse events. Overall, FMT appears to be safe and effective for treatment of infectious complications, including CDI and colonisation of multidrug-resistant bacteria.

	Indication or population	Number of patients	Administration route	Study type	Donor relation	Number of FMT procedures	Safety	Response
Kakihana et al (2016) ⁶⁶	Steroid-resistant or steroid-dependent gut GVHD	4	Nasogastric tube	Prospective	Spouse or relative	7	1 lower gastrointestinal bleeding, hypoxaemia, probably not related to FMT	3 CR, 1 PR
Spindelboeck et al (2017) ⁶⁷	Steroid-resistant, grade IV gut GVHD	3	Colonoscopy	Retrospective, case series	Unrelated or sibling	9	No serious adverse event	2 CR, 1 PR
Kaito et al (2018) ⁶⁸	Steroid-resistant, grade IV gut GVHD	1	Oral capsules	Case report	Sister	2	No serious adverse event	1 PR
Qi et al (2018) ⁶⁹	Steroid-resistant gut GVHD	8	Nasoduodenal tube	Prospective	Unrelated	12	No serious adverse event	5 CR, 1 PR
Shouval et al (2018) ⁷⁰	Steroid resistant/dependent gut GVHD	7	Oral capsules	Prospective	Unrelated	15	2 bacteraemia (deemed unrelated)	2 CR
Mao et al (2020) ⁷¹	Steroid-resistant gut GVHD	1	Oral capsules	Case report	Unrelated	2	No serious adverse event	1 CR
van Lier et al (2020) ⁷²	Steroid resistant or steroid-dependent GVHD	15	Naso-duodenal tube	Prospective	Unrelated	15	No serious adverse event	10 CR
Goeser et al (2021) ⁷³	Steroid-resistant gut GVHD	11	Nasoduodenal tube or capsule	Retrospective	Mostly unrelated	11	No serious adverse event	Stool frequency and volume, and aGVHD grading
Bilinski et al (2021) ⁶⁵	Steroid-resistant gut GVHD	11	Nasoduodenal tube	Retrospective	Unrelated	14	2 bacteraemia, 1 norovirus infection	4 CR, 1 PR
Malard et al (2021) ⁷⁴	Steroid-resistant GVHD	24	Enema	Prospective	Unrelated	24	4 infectious adverse event possibly related to FMT, 1 probably related to FMT	9 CR, 4 PR
Malard et al (2021) ⁷⁴	Steroid-resistant or steroid-dependent GVHD	52	Enema	Prospective	Unrelated	52	2 gastrointestinal symptoms, 6 bacterial infections, possibly related to FMT	21 CR, 14 PR

FMT=faecal microbiota transplantation. GVHD=graft-versus-host disease. CR=complete response. PR=partial response. aGVHD=acute GVHD.

Table 2: FMT studies in haematopoietic cell transplantation recipients for the treatment of GVHD

Results of FMT in allogeneic HCT recipients

Treatment of refractory acute GVHD

Kakihana and colleagues⁶⁶ reported on FMT for treatment of four patients with steroid-resistant (n=3) and steroid-dependent (n=1) gastrointestinal acute GVHD in 2016. FMT donors were either spouse or relative, and faecal suspension was administered via a nasoduodenal tube. After one or two FMT treatments were administered 1 week or 2 weeks apart, all patients were in response, including two complete response and one partial response; no serious adverse events related to the FMT were reported. On these basis of these promising results, several case reports and small series were published (table 2). In particular, van Lier and colleagues⁷² reported use of FMT in 15 patients with steroid-resistant (n=6) or steroid-dependent (n=9) gastrointestinal acute GVHD. All patients received a single FMT with fresh faeces delivered through a nasoduodenal tube. Of note, all patients but one received FMT from the same donor. In ten (67%) of the participants, a complete clinical response was observed within 1 month after FMT, including three (50%) responses in patients with steroid-resistant acute GVHD and seven (77%) responses in patients with steroid dependent acute GVHD. In four (40%) of ten patients in complete response, GVHD symptoms returned during immunosuppressive drug tapering. Achievement of a complete response without a secondary

failure was associated with an improved overall survival at 24 weeks, compared with non-responders ($p=0.0068$). Furthermore, response correlated with high gut microbial α -diversity, partial engraftment of donor bacterial species, and an increase in abundance of butyrate-producing bacteria, including Clostridiales and *Blautia* species.

Our team reported clinical outcomes from a cohort of 76 patients with steroid-refractory gastrointestinal-acute GVHD treated with a pooled allogeneic microbiome ecosystem therapy, MaaT013.⁷⁴ 24 patients were treated in the prospective, international, single-arm, phase 2a HERACLES study (NCT03359980) and 52 in a compassionate use and expanded access programme in France. MaaT013 is a product manufactured under current good manufacturing process guidelines with faecal material from strictly vetted, healthy donors (three to six donors) and characterised by a highly consistent richness of 455% (relative SD 3) operational taxonomic units. It is composed of 30 g of faeces frozen at -80°C , with a stability of 24 months. Treatment comprised three MaaT013 doses administered weekly by enema. All patients received at least one MaaT013 dose; 22 (92%; HERACLES) and 45 (87%; expanded access programme) received at least two doses, and 12 (50%; HERACLES) and 37 (71%; expanded access programme) received the full treatment course. In the prospective population, the gastrointestinal overall response rate was

38% (n=9) at day 28, including five complete responses, two very good partial responses, and two partial responses. In the expanded access programme, the gastrointestinal overall response rate was 58% (n=30) at day 28 (17 complete responses, nine very good partial responses, and four partial responses). Considering the best gastrointestinal overall response rate, 13 (54%) of 24 patients achieved at least a partial response in the prospective trial and 35 (67%) of 52 patients in the expanded access programme. The 12-month overall survival was higher in patients responding at day 28 than in non-responders in the prospective study (44% for responders vs 13% for non-responders, $p=0.047$) and in the expanded access programme (59% for responders vs 7% for non-responders, $p<0.0001$). In the prospective trial, only four possibly and one probably related serious treatment-emergent adverse events were reported, and all were infectious events. Nevertheless, shotgun sequencing analyses of the identified strains revealed that none were found in MaaT013. In this prospective trial, patients responding at day 28 had higher values of faecal bacterial microbiota α -diversity indices (Shannon $p=0.005$ and Richness $p=0.038$) than had non-responding patients, and showed MaaT013 engraftment with higher proportions of MaaT013-derived species in the total composition of microbiota ($p=0.04$).

Perspectives of microbiota modulation in allogeneic HCT

Beyond the use of FMT for treatment of CDI or eradication of multidrug-resistant bacteria or gastrointestinal acute GVHD, prophylactic FMT might be effective in preserving gut microbiota composition and improve patients' outcomes. Taur and colleagues⁷⁵ reported the results of a phase 2 randomised controlled trial of autologous FMT versus no intervention after allogeneic HCT. Stools were collected before administration of the transplantation-conditioning regimen and stored frozen (-80°C). Faecal samples were analysed and patients whose pre-allogeneic HCT faeces showed low microbial diversity or tested positive for the presence of intestinal pathogens were excluded. Results of the first 25 patients (14 who received autologous FMT treatment and 11 patients in the control group who did not) revealed that the autologous FMT intervention boosted microbial diversity and re-established the intestinal microbiota composition that existed before the patient had received antibiotic treatment and allogeneic HCT. In addition, autologous FMT was associated with increased neutrophil, lymphocyte, and monocyte counts compared with patients in the control group.⁷⁶

Another study enrolled 18 patients who were to receive frozen FMT capsules from unrelated donors for gut microbiota restoration after allogeneic HCT.⁷⁷ Five patients did not receive FMT because of the development of early gastrointestinal acute GVHD before FMT (n=3), persistent gastrointestinal toxicity

related to allogeneic HCT (n=1), or patient decision (n=1). 13 patients received 15 FMT capsules per day for two consecutive days, up to a total of 30 capsules, at a median of 27 days (range 19–45) after allogeneic HCT. Patients were able to swallow and tolerate all FMT capsules. FMT was tolerated well, with one treatment-related serious adverse event (grade 3 abdominal pain). Two patients subsequently developed gastrointestinal acute GVHD, with one patient also having concurrent bacteraemia; no additional cases of bacteraemia occurred. Analysis of stool composition indicated improvement in intestinal microbiome diversity after FMT, which was associated with expansion of donor-stool taxa.

In a phase 2, single-arm, multicentre trial, we studied autologous FMT in 25 patients with acute myeloid leukaemia treated with intensive chemotherapy and antibiotics.⁸ Faeces were collected and frozen (-80°C) at diagnosis of acute myeloid leukaemia before patients received any chemotherapy or antibiotic treatment. We found that intensive induction chemotherapy and antibiotics induced a dramatic decrease in α -diversity indices and microbial dysbiosis with a substantial shift of the microbial communities and domination of proinflammatory families. After autologous FMT treatment, α -diversity indices returned to their initial mean values and the similarity index showed the restoration of microbial communities. Furthermore, although there was an increase in antibiotic resistance genes after induction chemotherapy, it decreased after autologous FMT. No serious adverse event related to the autologous FMT was reported. 18 (72%) patients received an allogeneic HCT and only three (17%) developed gastrointestinal GVHD, suggesting a possible protective effect of FMT after induction chemotherapy on patients' outcomes after allogeneic HCT.⁸

Based on these results and given that large multicentre studies have confirmed that higher versus to lower diversity of intestinal microbiota at engraftment (and also before starting conditioning) was associated with a lower risk of death,¹³ FMT is a promising treatment to restore gut microbiota diversity before and after allogeneic HCT and improve patients' outcomes.

Perspectives of microbiota modulation in patients with haematological malignancies beyond allogeneic HCT

Similar to allogeneic HCT, given the impact of microbiota and use of antibiotics on patients' outcomes after autologous HCT²³ and CAR T-cells,²⁶ FMT might restore gut microbiota diversity in these patients, not only to decrease toxicity of the procedure, but also to decrease risk of relapse and improve outcomes. Nevertheless, before it can be used routinely, large multicentre studies are needed to confirm the findings, particularly with CAR T cells, to understand the underlying mechanism and to evaluate safety and efficacy. The effect of antibiotics on the gut microbiota

with other CAR T-cell treatments, such as anti-BCMA CAR T cells that are already approved for multiple myeloma and other CAR T cell-based candidates in development, should be evaluated. Gut microbiota manipulation with FMT might also be an effective adjuvant therapy in these patients after autologous HCT²³ and CAR T-cells, provided the possible role of the gut microbiota on progression and development and response to treatment of myeloma^{28,29} and lymphoma³⁰ is confirmed in larger multicentre studies.

Conclusion

Presently, the association between the gut microbiota and patient outcome has been established for allogeneic HCT recipients and also autologous HCT and CAR T cell recipients. Furthermore, preliminary data suggest a link between multiple myeloma and lymphoma history and response to treatment. Thus, gut microbiota modulation appears to be the next cornerstone in improving outcomes in patients with haematological malignancies. Several approaches are possible, including dietary change and administration of either selected probiotics, whole microbiome with FMT, or postbiotics. FMT is the most advanced approach, with first evidence that it is an effective adjuvant therapy in the setting of solid cancers.^{78,79} In the setting of haematological malignancies, the exact immune modulation effect of FMT remains to be established. Nevertheless, autologous FMT was shown to be associated with increased neutrophil, lymphocyte, and monocyte counts compared with patients in the control group.⁷⁶ In addition, Andrlová and colleagues⁸⁰ have reported that a diverse intestinal microbiome at engraftment after allogeneic HCT is associated with an increased number of innate-like mucosal-associated invariant T (MAIT) cells, which are, in turn, associated with improved overall survival and less acute GVHD. Similarly, Vδ2 cell frequencies (the major circulating subpopulation of γδ T cells) correlated with intestinal microbial diversity and predicted acute GVHD. Although the effect of FMT on MAIT and Vδ2 cell recovery remains to be explored, this correlation suggests that restoration of a diverse intestinal microbiome with FMT can modulate MAIT and Vδ2 through the production of

activating ligands. We believe gut microbiota modulation will probably be the next revolution in the management of cancer, including haematological malignancies.

Contributors

All authors contributed substantially to the conception, writing, critical review, and final approval of the manuscript.

Declaration of interests

FM reports honoraria from Therakos (Mallinckrodt), Janssen, Biocodex, Sanofi, JAZZ Pharmaceuticals, Gilead, Novartis, and Astellas, all outside the submitted work. MM reports grants and honoraria from Janssen, Sanofi, and JAZZ Pharmaceuticals; honoraria from Celgene, Amgen, MaaT Pharma, Bristol Myers Squibb, Takeda, and Pfizer; and grants from Roche, all outside the submitted work. BG has no competing interests.

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Search strategy and selection criteria

We searched PubMed, Embase, and the Cochrane Library for articles and reviews published in English between Jan 1, 2018, and May 1, 2022, and older references were also used when appropriate. We used the search terms “microbiota” and “hematologic malignancies” or “allogeneic hematopoietic cell transplantation”. We reviewed the abstracts and retrieved the full texts of articles that were deemed relevant. We also searched the reference lists from articles and reviews identified by the search. The final reference list was generated on the basis of originality and relevance to the scope of this Review.

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